# FUTURE-SPLITTING STATE RECRUITMENT IN HISTORY-DEPENDENT HCP MAGNESIUM MECHANICS

*A Minimum-Decisive Prospective Protocol for State Sufficiency, Probe Selection, and Mechanism Localisation*

**Mark Charles McLaughlin**

Independent Researcher | ORCID 0009-0005-7736-1511

Version 1.0 | 4 September 2026

Reserved DOI: 10.5281/zenodo.22309144

> **PROSPECTIVE PROTOCOL.** This paper reports a predeclared experimental design. It reports no experimental confirmation of MKUFT, FSAI, or FSSR, and introduces no new constitutive law, twinning mechanism, force, field, or completed unification claim.

Framework lineage: McLaughlin-Kairos Unified Field Theory (MKUFT) / Future-Sufficient Address Invariant (FSAI) / Future-Splitting State Recruitment (FSSR)

## Abstract

History-dependent constitutive mechanics provide a direct test of whether a state representation is sufficient for the future it is asked to predict. This paper specifies a minimum-decisive prospective protocol for the Future-Sufficient Address Invariant / Future-Splitting State Recruitment (FSAI/FSSR) family in HCP magnesium, using AZ31/AZ31B as the default reference system. The protocol does not test the trivial proposition that magnesium exhibits twinning, detwinning, hysteresis, or loading-history effects; those are established materials-science objects. Instead, it starts from the strongest practical predeclared domain-native constitutive state and asks whether a lawful reversal chosen specifically to expose false state equivalence can produce a held-out future split beyond a frozen tolerance. If so, the protocol requires exhaustion of state-matching, challenge, boundary, measurement, stochastic, and probe-induction nulls before recruiting the smallest predeclared measurable physical augmentation that restores held-out closure. A remove/restore test then asks whether predictive failure reopens when that augmentation is removed and contracts when it is restored. In parallel, the FSSR challenge is compared under matched budget with a parameter-information challenge selected by standard experimental-design logic. State sufficiency, challenge-design value, mechanism localisation, and prospective lead relative to a conventional macroscopic marker are reported as four independent verdicts. The experiment is explicitly allowed to return preservation or null results. The design is therefore a preregistration-ready attempt to make one central MKUFT state-adequacy claim prospectively killable in a native materials system.


**Keywords:** Future-Splitting State Recruitment; FSSR; Future-Sufficient Address Invariant; FSAI; HCP magnesium; AZ31; AZ31B; history dependence; constitutive modelling; state sufficiency; twinning-detwinning; crystal plasticity; experimental design; active probing; mechanism localisation; predictive closure; preregistration.

> *Core discriminator: The flagship is not “magnesium has memory.” It is whether a prospectively chosen future can expose exactly where a strong state description stops being enough, whether the smallest physical repair makes it enough again, and whether that way of asking the question beats the strongest ordinary way of asking it.*

## 1. Scope, status, and lineage

This paper freezes the first minimum-decisive physical/material flagship protocol for the FSAI/FSSR family. Its scientific object is a state-adequacy test, not a claim to ownership of established magnesium mechanics. The live MKUFT module from which this paper is derived is Module 28C; this PDF is the frozen publication carrier associated with reserved DOI 10.5281/zenodo.22309144.

The immediate methodological lineage is Future-Splitting State Recruitment (FSSR) [6], Addressed Admissible Futures (AAF) [7], the Future-Sufficient Address / preserve-or-reopen compositional work [8], and the current principal MKUFT architecture [9]. This lineage supplies the state, future, readdressing, experimental-governance, and falsification language. It does not turn those framework relations into materials-science evidence.

### 1.1 Exact primary claim

> *A strong reduced constitutive state that is adequate in a baseline regime can become prospectively inadequate under a lawful separating reversal; an FSSR-selected challenge can expose that loss of state sufficiency, and the smallest predeclared measurable physical augmentation can restore held-out predictive closure.*

Three stronger claims are scored separately: (i) challenge-design value relative to a matched parameter-information design, (ii) physical mechanism localisation of the recruited repair, and (iii) prospective lead where recruitment becomes necessary before a conventional macroscopic constitutive marker is overt. No one claim inherits success from another.

### 1.2 What success would not establish

- MKUFT did not discover twinning, detwinning, cyclic hysteresis, strain localisation, or internal-variable constitutive modelling.
- A positive result would not show that HCP magnesium requires a new physical law, that information is an autonomous force, or that S-I-P-O is a new spatial dimension.
- The protocol does not supersede crystal plasticity, constitutive modelling, active learning, optimal experimental design, observability, state estimation, or hysteresis theory.

- One positive materials result would not validate MKUFT claims in biology, AI, quantum foundations, gravity, consciousness, or metaphysics.
- No outcome of this protocol alone establishes a completed unified field theory.

## 2. Native system and prior-art boundary

Use a textured HCP magnesium alloy with a mature cyclic/reversal literature, with AZ31/AZ31B as the default reference system unless the executing laboratory has a better-characterised equivalent. The selection is practical rather than proprietary.

Existing work already supplies the native phenomena, models, and measurement routes needed to make the test non-trivial. Constitutive twinning-detwinning models exist for HCP polycrystals [1]; cyclic AZ31 experiments show macroscopic twin bands and strain localisation under DIC [2]; multiscale crystal-plasticity work has been validated against in-situ synchrotron XRD and SEM-EBSD [3]; in-situ neutron diffraction, identical-area EBSD, and TEM have directly tracked twinning/detwinning in AZ31 [4]; and cyclic bending has been studied using in-situ DIC, EBSD, and EVPSC twinning-detwinning modelling [5].

> *Prior-art boundary: these papers are native ownership anchors and feasibility evidence. They are not evidence for MKUFT. The MKUFT-specific delta exists only after the strongest fair domain-native explanation has been subtracted.*

## 3. Declared experimental address

Before confirmatory data collection, freeze the complete target-relevant experimental address. A nominally identical reversal is not automatically the same challenge: any coordinate capable of changing the declared future belongs to the challenge or boundary.

- material, batch, texture, and specimen geometry
- loading axis, fixture, control mode, and thermal regime
- strain/stress rate, dwell, preload, alignment, and acquisition timing
- history-generation family and candidate state representation Theta_0
- future target q, prediction horizon Delta, challenge family U
- boundary/environment class E and registration family O
- trajectory discrepancy d and closure tolerance epsilon_q
- candidate repair library C and ordinary model/null family M_0
- matched experimental budget for challenge-design comparison

### 3.1 Primary future object

The principal claim uses one future object: the reverse-loading macroscopic stress trajectory over a preregistered strain window after the challenged state.

$$q = \sigma_{\mathrm{rev}}(\varepsilon)$$
$$\mathbf{y} = [\sigma(\varepsilon_1), \ldots, \sigma(\varepsilon_m)]$$

A default trajectory discrepancy is the covariance-standardised distance below. Sigma_0 is estimated only from calibration/development data and frozen before confirmation; a laboratory may replace this with another prospectively justified trajectory metric, but may not choose the metric after seeing the confirmatory split.

$$d_q(\mathbf{y}, \mathbf{y}') = \sqrt{\frac{1}{m}(\mathbf{y} - \mathbf{y}')^T \Sigma_0^{-1} (\mathbf{y} - \mathbf{y}')}$$

Set epsilon_q from baseline replicate variability plus a prospectively declared smallest material difference of interest. Do not define the tolerance from the confirmatory cross-history split.

Secondary readouts may include twin volume fraction, variant distribution, texture evolution, DIC/HRDIC strain fields, diffraction/lattice-strain observables, slip/twin activity summaries from a separately validated model, hardening/back-stress proxies, and twin-band/localisation descriptors. These readouts localise mechanism and test repair; they do not replace the frozen primary target after the result is known.

## 4. State hierarchy: no straw-man baseline

A deliberately coarse macroscopic state may be used only as a calibration surface. A history split under such a state is expected in many cyclic magnesium regimes and is not the flagship result.

The confirmatory object Theta_0 is the strongest practical reduced domain-native constitutive state selected before confirmatory data. It must include every state variable that the chosen serious ordinary model normally requires and that is available under the agreed measurement budget. Residual twin fraction, hardening state, texture, or other established internal coordinates may not be artificially omitted to manufacture success.

Where feasible, also include a stronger history-aware constitutive baseline, a flexible nonparametric or machine-learned history model, and a high-capacity latent-state model evaluated on held-out prediction without interpreting arbitrary latent coordinates as mechanisms. If one of these closes the declared future more simply and equally well, the MKUFT-specific claim contracts.

The model whose state is being tested cannot certify itself. Confirmatory continuation must come from physical experiment or a separately validated higher-fidelity microstructure-resolved model; for the strongest claim, physical experiment is decisive and simulation is a localisation/deformation aid rather than the sole judge.

## 5. Controlled history construction

Construct a bounded family of loading histories capable of reaching states that are near-matched under Theta_0 while retaining different microstructural histories. A useful development family may contrast a compression-dominant prehistory followed by partial reversal/detwinning against a slip-dominant or differently ordered reversal history reaching the same target state. Exact paths are development objects; the confirmatory pair-generation rule is frozen before Stage 2.

$$D_{\Theta}\left(\Theta_0(h), \Theta_0(h^{'})\right) \leq \tau_{\Theta}$$

The matching metric and tau_Theta are frozen from development/calibration variability and model sensitivity. The state-matching audit must explicitly test whether any future split is better explained by residual mismatch already visible inside Theta_0. Individual specimens may not be cherry-picked after their futures are inspected.

## 6. Challenge selection: the question behind the question

FSSR does not merely ask for a strong reversal. It asks which lawful future best tests whether Theta_0 is really a state for q. Create a bounded challenge family U by varying only a small number of native controls such as reversal direction, reversal amplitude, bounded dwell, and one preregistered loading-rate level if scientifically required. Keep unrelated conditions fixed.

### 6.1 FSSR state-splitting challenge

$$u_F \in \arg\max_{u \in \mathcal{U}} J_{\text{split}}(u; \Theta_0)$$

On development data only, J_split rewards future separation inside matched Theta_0 fibres while penalising measurement instability and challenge-induced mismatch. Freeze u_F before confirmatory continuation.

### 6.2 Parameter-information comparator

Using the same development data, challenge family, information access, and experimental budget, select u_PI using a standard parameter-identification or information objective appropriate to the executing model, such as Fisher information, posterior-entropy reduction, or active-learning uncertainty reduction. Optionally add one ordinary standard-practice reversal u_STD selected before confirmatory outcomes are inspected.

If u_F and u_PI are equivalent under fair tuning and yield equivalent held-out state discrimination, the separate FSSR challenge-selection claim is NULL even if a genuine state-recruitment event remains.

## 7. Native consequence cone

Stress the state-sufficiency relation inside the smallest target-sufficient native consequence surface. The surface should contain enough native structure for predicted downstream deformation, compensation, and restoration to appear, but not so much uncontrolled ecology that ownership becomes unidentifiable.

$$K_q(r \mid a, E, \mathcal{U}, \Delta)$$

For this protocol, the candidate cone includes only the prehistory/current Theta_0 state, reversal challenge and fixture/boundary state, macroscopic force/stress/strain trajectory, the microstructural twin/slip state needed for repair testing, local strain/texture readouts when they test compensation or localisation, ordinary redundant slip/twin routes, and the recovery/reversal segment needed to test restoration.

A macroscopic-only thin control may show that paths differ but cannot receive mechanism-localisation credit if it cannot observe which native relation carried the divergence. Conversely, uncontrolled additions of temperature, geometry, rate, multiaxiality, or material variation are not stronger merely because they are more realistic. The cone is sufficient, not maximal.

## 8. Candidate repair library and minimum-repair rule

After development and before confirmation, predeclare a physically measurable candidate repair library C, excluding coordinates already present in Theta_0. A useful nested hierarchy is:

- C1 - total/residual twin volume fraction, if not already owned by Theta_0;
- C2 - variant-resolved twin fraction or orientation distribution;
- C3 - low-dimensional texture or twin-topology descriptor;
- C4 - slip/twin activity or internal/back-stress summary with independent physical interpretation;
- C5 - local strain-heterogeneity or twin-band descriptor from DIC/HRDIC;
- C6 - bounded physically interpretable history functional.

The executing materials team may replace this hierarchy with a stronger domain-native predeclared library. Arbitrary latent neurons or coordinates may improve prediction but do not become physical mechanisms by naming.

Prefer the simplest successful repair by the prospectively declared order: independently measurable physical descriptor; lower dimension; lower acquisition burden; narrower dependency cone; equal or better held-out closure. A coalition is allowed only if every simpler predeclared candidate fails or domain theory prospectively requires the coalition.

## 9. Development, freeze, confirmation, and replication

### 9.1 Stage 0 - apparatus and native calibration

- repeatable stress/strain acquisition and stable boundary/alignment/rate/thermal conditions;
- reliability of DIC, EBSD, XRD, neutron, or other selected microstructure readouts;
- known twinning/detwinning response in the selected orientation/loading regime;
- injected or synthetic analysis checks where appropriate;
- baseline replicate distributions needed for Sigma_0, tau_Theta, and epsilon_q.

Stage 0 earns instrument competence only.

### 9.2 Stage 1 - development / identification

- choose and freeze Theta_0 and serious ordinary baselines;
- define the controlled history-pair generator and challenge family U;

- select u_F, u_PI, and optional u_STD;
- define the candidate repair library and simplicity order;
- set the primary trajectory discrepancy and tolerance;
- set the conventional macroscopic transition marker lambda_T;
- estimate variance and prospective precision/power;
- freeze exclusion, stopping, QC, and missing-data rules.

Stage 1 cannot confirm any quantity it selected.

### 9.3 Preregistration freeze

Before Stage 2, commit or externally preregister the material/batch inclusion rules, history-pair generator, Theta_0 and comparators, U/u_F/u_PI/u_STD, q/Delta/d/epsilon_q/tau_Theta, candidate repair library and simplicity order, primary and secondary endpoints, lambda_T definition, sample-size/precision calculation, exclusions/stopping/missing-data treatment, analysis code or immutable hash, and the success/null/contraction decision table.

### 9.4 Stage 2 - held-out confirmation

1. Do histories matched under frozen Theta_0 split beyond epsilon_q under u_F?
2. Does the strongest state/challenge/boundary/measurement/stochastic null explain the split?
3. Does one frozen physical augmentation c* restore held-out closure?
4. Does removing c* from the repaired predictive state reopen the held-out failure?
5. Does u_F expose state inadequacy more strongly or efficiently than u_PI under matched budget?

### 9.5 Stage 3 - native deformation / mechanism localisation

Use the strongest lawful combination of a separately validated CPFE/EVPSC or equivalent high-fidelity model with targeted internal-state deformation; prospectively selected physical history interventions that vary the candidate microstructural coordinate while matching the rest of Theta_0 as closely as the system permits; compensation controls that vary the strongest alternative slip/twin route; and restoration/reversal tests that predict contraction of the same deformation signature.

> *Computational deletion is computational evidence. It is not physical ablation. If exact physical isolation of c* is infeasible, mechanism ownership remains bounded or unresolved rather than being declared by model convenience.*

### 9.6 Stage 4 - independent replication

- fresh material batch
- different operator or team
- blinded or condition-masked confirmatory analysis
- independently repeated loading series
- second texture/orientation or closely related HCP material if generality is claimed
- adversarial review of the strongest native ordinary model

## 10. Empirical residual and predictive closure

For matched histories under a frozen challenge, estimate the future-splitting residual with the preregistered trajectory discrepancy. UpperTail_pre is fixed before confirmation, for example as a specified high quantile with uncertainty or another robust worst-case estimator justified by the sample design.

$$\widehat{R}_q(\Theta_0; u) = \mathrm{UpperTail}_{\mathrm{pre}}\{d_q(\mathbf{y}_h, \mathbf{y}_{h'}) : D_\Theta(\Theta_0(h), \Theta_0(h')) \le \tau_\Theta\}$$

State insufficiency for the declared experiment requires:

$$\widehat{R}_q(\Theta_0; u_F) > \varepsilon_q$$

A recruitment candidate c closes only when the held-out repaired state satisfies:

$$\widehat{R}_q(\Theta_0 \oplus c; u_F) \leq \varepsilon_q$$

The predictive remove/restore test is:

> $\Theta_0$ + c* closes
> remove c* with the frozen repaired model otherwise unchanged → failure reappears beyond tolerance
> restore c* → closure returns on an independent or held-out continuation

This is a predictive-state deformation test. Physical mechanism ownership additionally requires Stage-3 native deformation.

## 11. Ordered deformation signature and probe-induction control

Before Stage 2, declare an ordered signature rather than merely predicting a larger error:

- matched Theta_0 histories show no primary difference beyond baseline tolerance before the separating reversal;
- u_F enters the future-splitting regime;
- primary reverse-loading trajectories diverge beyond epsilon_q;
- one or more predeclared microstructural readouts diverge in an order predicted by the native materials model;
- adding c* restores held-out trajectory closure;
- removing c* reopens the trajectory failure;
- targeted restoration/deformation contracts or reproduces the same ordered signature.

The separating challenge may itself create twinning or another distinction rather than reveal a distinction already carrying different futures. Stage 1 therefore defines at least one probe-induction control: a lower-amplitude nested challenge, a passive/pre-challenge microstructure measurement, a challenge series identifying whether divergence existed before the overt transition, a validated high-fidelity continuation showing the candidate coordinate was already different, or another domain-native method separating revelation from induction. If the split vanishes when induction is controlled, the hidden-state claim contracts to a challenge-created transition.

## 12. Conventional macroscopic marker and recruitment lead

Define the conventional marker from a macroscopic constitutive criterion before Stage 2, not from the candidate repair. A default form is the earliest loading coordinate at which the macroscopic trajectory departs from a preregistered baseline/slip-dominant constitutive envelope by more than epsilon_T for a required number of consecutive increments.

$$L = \lambda_T - \lambda_R$$

lambda_R is the earliest prospectively confirmed loading coordinate at which Theta_0 requires c* to preserve the declared future within epsilon_q. A positive lead claim requires L > 0 with preregistered uncertainty. Overlapping intervals or loss of lead under replication returns a NULL prospective-lead verdict even if state recruitment remains valid.

## 13. Strongest fair null family

| Null | Attack | Required contraction |
|---|---|---|
| N1 State mismatch | Tighter matching or a variable already owned by Theta_0 removes the split. | Ordinary matching failure; no FSSR recruitment credit. |
| N2 Stronger native state | A serious domain-native constitutive model already closes the future. | Native model wins; MKUFT delta contracts or becomes translation only. |

| Null | Attack | Required contraction |
|---|---|---|
| N3 Flexible history model | A matched-information flexible history-aware model closes held-out futures equally or better. | FSSR must still earn minimality, interpretability, probe efficiency, or preservation/reopening; otherwise NULL. |
| N4 Challenge/boundary mismatch | Amplitude, rate, dwell, fixture, thermal, alignment, or related differences carry the split. | Readdress challenge/boundary; no hidden-state promotion. |
| N5 Measurement artefact | Acquisition, registration, interpolation, or analysis creates the apparent divergence. | No state failure until the instrument null is removed. |
| N6 Ordinary stochastic spread | The split lies inside the frozen variability model. | Preserve Theta_0 for the declared scope. |
| N7 Probe induction | u_F creates the candidate mechanism rather than exposing a pre-existing future distinction. | State-recruitment interpretation contracts. |
| N8 Parameter-design equivalence | u_PI selects an equivalent challenge and equal held-out discrimination. | FSSR challenge-design delta is NULL; state result scored separately. |
| N9 Arbitrary latent repair | Only a high-dimensional latent coordinate closes, without stable physical interpretation. | Predictive repair may stand; mechanism localisation fails. |
| N10 Repair overfit | c* closes development but not fresh confirmation or remove/restore. | Recruitment fails. |
| N11 Generic damage | Comparable disturbance produces the same trajectory degradation. | Relation-specific ownership fails. |
| N12 Compensation | A redundant slip/twin route preserves the parent trajectory despite local deformation. | Local load may be real while parent closure remains intact. |

## 14. Four independent verdicts

| Verdict family | Class | Meaning |
|---|---|---|
| V1 State sufficiency | PRESERVE | Strong separating challenge; no material held-out split. |
| | REOPEN | Frozen matched state splits beyond epsilon_q after null audit. |
| | FAIL-TO-LOCALISE | Split exists but no frozen minimal repair restores held-out closure. |
| V2 Challenge design | FSSR-GAIN | u_F gives stronger or cheaper state-adequacy discrimination than u_PI under matched budget. |
| | EQUIVALENT / NULL | u_F and u_PI are materially equivalent. |
| | WORSE | u_PI is prospectively superior. |
| V3 Mechanism localisation | PHYSICALLY TYPED | c* is measurable, held-out, deformation-sensitive, and survives native nulls. |
| | PREDICTIVE ONLY | c* improves prediction but physical ownership remains unresolved. |
| | NULL / FAIL | No stable repair, or ordinary model fully owns the effect. |
| V4 Prospective lead | LEAD | lambda_R < lambda_T prospectively with declared precision. |
| | SIMULTANEOUS / NULL | No reproducible lead. |
| | REVERSED | Conventional marker becomes informative first. |

> *No celebratory scalar: success in V1 cannot be used to fabricate success in V2, V3, or V4.*

## 15. Decision table

| Confirmatory pattern | State verdict | Interpretation |
|---|---|---|
| No split under strongest lawful u_F | Preserve Theta_0 | Clean null; simpler state survives. |
| Split disappears with tighter matching/boundary control | No state failure | Misaddressed state or challenge. |
| Split survives; established ordinary variable closes | Repair valid | Ordinary state reconstruction; MKUFT novelty NULL unless challenge/minimality adds prospective value. |
| Split survives; c* closes held-out; u_PI matches u_F | Recruitment supported | Challenge-objective delta NULL. |
| Split survives; c* closes; remove reopens; u_F beats u_PI | Strong methodological support | FSAI/FSSR procedure adds prospective value in this domain. |
| Same + independent physical deformation/replication + L>0 | Tier-3 candidate result | Bounded mechanism-level flagship support only. |
| Repair fails held-out or loses to ordinary baseline | Recruitment fails | Contract branch; do not add hidden state post hoc. |

## 16. Preregistration, power, and analysis custody

This protocol does not guess a universal specimen count. Stage 0/1 estimates variance and repeatability; Stage 2 sample size follows a prospective precision/power calculation tied to the frozen epsilon_q, smallest effect of interest, and the hierarchical structure of specimens, grains, and measurements.

- specimen-level independence respected in inference; no pseudoreplication from thousands of pixels or grains;
- immutable raw files and acquisition metadata;
- condition labels masked from the confirmatory analyst where feasible;
- frozen preprocessing and multiplicity correction for secondary candidate repairs;
- one primary state-sufficiency endpoint;
- null publication;
- public code and data where legally and practically possible.

If a candidate repair is selected from Stage-2 outcomes, it becomes a new discovery object and requires fresh confirmation. Stage 2 cannot both invent and confirm it.

## 17. Native-consequence-cone acceptance and falsification

Relation-specific credit is earned only if the stress surface is rich enough to show the relevant downstream signature and narrow enough to retain attribution. Before accepting a positive relation-deformation result, ask whether the local microstructural relation changed first as predicted, whether compensation behaved as predicted, whether the macroscopic future moved only when the declared parent relation should move, whether restoration contracted the same signature, whether a thin fixture missed information the native cone exposed, and whether a richer uncontrolled environment added anything except attribution noise.

If the native cone adds no discrimination over the reduced experiment, prefer the reduced experiment and contract the consequence-cone claim for this target.

## 18. Closure, reopening, and interpretation boundary

Design closure is earned when the exact claims and non-claims are explicit; native domain ownership is explicit; the primary target and state are frozen in form; the fair-null family is specified; FSSR and parameter-information challenges face matched budgets; candidate-repair minimality is prospectively governed; consequence-cone, deformation, restoration, and replication burdens are explicit; and every positive branch has a corresponding contraction/null route.

Empirical status remains OPEN until an external or independently executed experiment passes the applicable stages. Repository completeness is not experimental evidence. A later execution must create a new result object containing the exact preregistration, data provenance, code/hash, outcomes, null audit, deviations, and four independent verdicts. This design paper must not be silently rewritten to make the observed result look predicted.

## 19. Canonical compression

> strong native state $\Theta_0$
> → construct near-matched histories without seeing confirmatory futures
> → choose u_F to attack false state equivalence
> → choose u_PI under the same budget to attack parameter uncertainty
> → freeze both
> → continue independently inside the smallest native consequence cone
> → split beyond tolerance OR preserve the state
> → exhaust state/challenge/boundary/measurement/stochastic/probe nulls
> → recruit the smallest predeclared measurable repair c*
> → test held-out closure
> → remove c* and demand reopening; restore and demand contraction
> → compare u_F against u_PI
> → localise physically only as far as native deformation allows
> → compare λ_R with frozen macroscopic marker λ_T
> → return four separate verdicts
> → replicate before Tier-3 promotion

> *The experiment is successful as a discriminator even when the answer is NULL. A strong native state that survives a strong separating challenge has earned preservation for the declared target, regime, horizon, challenge family, and tolerance.*

## Publication identity and framework lineage

This PDF is the frozen v1.0 publication carrier prepared for Zenodo deposit under reserved DOI 10.5281/zenodo.22309144. The live GitHub module remains a later-editable research object; it must not silently rewrite the deposited version.

| Object | Title | Version | DOI | Relation |
|---|---|---|---|---|
| This paper | Future-Splitting State Recruitment in History-Dependent HCP Magnesium Mechanics | v1.0 | 10.5281/zenodo.22309144 | Reserved / prospective protocol |
| FSSR | Future-Splitting State Recruitment: A Cross-Domain Assay for State Adequacy and Prospective Mechanism Activation | v1.0 | 10.5281/zenodo.22058303 | Direct methodological parent |
| AAF | Addressed Admissible Futures: Future-Sufficient State, Load-Bearing Relations, and Restorative Reachability | v0.1 | 10.5281/zenodo.22031333 | Future/admissibility parent |
| Compositional schema | Cross-Domain Compositional Schema: Future-Sufficient Interfaces, Load-Bearing Relations, and Preserve-or-Reopen Reuse | v0.4 | 10.5281/zenodo.22166468 | Future-sufficient interface / preserve-or-reopen lineage |
| MKUFT | MKUFT - A Relational Architecture for Physical Law and Cross-Scale Dynamics | v2 | 10.5281/zenodo.21973064 | Principal framework publication |

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## Recommended citation

McLaughlin, M.C. (2026). Future-Splitting State Recruitment in History-Dependent HCP Magnesium Mechanics: A Minimum-Decisive Prospective Protocol for State Sufficiency, Probe Selection, and Mechanism Localisation (Version 1.0). Zenodo. https://doi.org/10.5281/zenodo.22309144

End of frozen v1.0 publication carrier | Reserved DOI 10.5281/zenodo.22309144